SHOPSPHERE EC2 RESTART & RECOVERY

Step-by-step runbook: what persists, what to restart, and how to verify the full setup

This guide is for restarting the existing stopped EC2 instance—not rebuilding it. Follow the steps in order. Commands are labelled by the terminal where they must run.

Known project setting	Value
AWS region	ap-southeast-2 (Sydney)
EC2 instance ID	i-0af087445cc5f5878
Instance type / OS	t3.small / Ubuntu 24.04
Deployment directory	/opt/shopsphere
Known-good application release	build-24
ECR registry	507941515612.dkr.ecr.ap-southeast-2.amazonaws.com
Monitoring on Windows	Prometheus 9091; Grafana 3000
EC2 metrics exporter	Node Exporter 9100 (private via SSM tunnel)
Jenkins on Windows	http://localhost:8080 (port 9090 is separately noted as used by Jenkins in monitoring setup)

Important: Stopping an EC2 instance does not normally erase its EBS root disk, installed software, Docker configuration, Compose files, image-tag.env, application .env, or Node Exporter service. It does stop the VM and its running processes. You must start the instance and re-establish sessions/tunnels. A normal public IPv4 address may change after start.

1. What persists and what must be restarted

Item	After EC2 Stop → Start
Ubuntu files and installed packages	Persist on the EBS root volume, unless the volume was deleted/replaced.
Terraform-managed infrastructure	Remains provisioned; starting the instance does not rerun Terraform.
Docker images cached on EC2	Usually remain on the root EBS disk.
Compose files, Nginx config, .env, image-tag.env	Remain on disk; do not overwrite or recreate them without checking.
Docker containers	Container definitions and writable layers persist, but running state stops with the VM; Docker restart policy should bri
Node Exporter package/service	Installed service remains; systemd enablement should start it on boot.
EC2 public IPv4 address	May change unless using an Elastic IP.
SSM interactive session / port-forward tunnel	Does not persist; start a new session.
Prometheus/Grafana on Windows	Unaffected by stopping EC2, but their PowerShell server processes must be running.
Prometheus target health	Will be DOWN while EC2/tunnel is stopped; should recover when tunnel and exporter are available.
ECR images	Remain in ECR independently of EC2; storage charges may continue.

Do not run Terraform destroy, delete the EBS volume, recreate the instance, or remove ECR repositories as part of an ordinary restart.

2. Start the existing EC2 instance

AWS Console

Open AWS Console → EC2 → Instances. Select instance i-0af087445cc5f5878 in ap-southeast-2 and choose Instance state → Start instance. Wait until Instance state is Running and both status checks pass (2/2).

Optional: start using AWS CLI from Windows PowerShell

```
aws ec2 start-instances --instance-ids i-0af087445cc5f5878 --region ap-southeast-2
aws ec2 wait instance-running --instance-ids i-0af087445cc5f5878 --region ap-southeast-2
aws ec2 describe-instance-status --instance-ids i-0af087445cc5f5878 --region ap-southeast-2
  --include-all-instances --query
  "InstanceStatuses[0].{State:InstanceState.Name,System:SystemStatus.Status,Instance:InstanceStatus.Status}"
  --output table
```

What: Starts the instance, waits for the Running state, then displays system and instance status checks. If AWS CLI is not configured for an identity with EC2 permissions, use the Console.

Check the public IP

```
aws ec2 describe-instances --instance-ids i-0af087445cc5f5878 --region ap-southeast-2 --query
  "Reservations[0].Instances[0].{State:State.Name,PublicIP:PublicIpAddress,PrivateIP:PrivateIpAddress}"
  --output table
```

Record the current public IP. Do not assume it is still 13.239.24.127. SSM commands use the instance ID, so the public IP is not needed for SSM access.

3. Wait for SSM to show the instance Online

SSM Agent must reconnect after the VM boots. Wait a few minutes after the EC2 status checks pass.

```
aws ssm describe-instance-information --region ap-southeast-2 --filters
  "Key=InstanceIds,Values=i-0af087445cc5f5878" --query
  "InstanceInformationList[0].{PingStatus:PingStatus,Platform:PlatformName,AgentVersion:AgentVersion}"
  --output table
```

Expected: PingStatus Online. If no result appears yet, wait 1–3 minutes and retry.

Open an interactive session (Windows PowerShell)

```
aws ssm start-session --target i-0af087445cc5f5878 --profile terraform --region ap-southeast-2
```

This opens the Ubuntu SSM terminal. If your active AWS profile differs, use the profile that has SSM permissions. Do not paste AWS secret keys into the EC2 terminal.

4. Verify Docker and the ShopSphere application

Run the following commands inside the Ubuntu EC2 SSM terminal.

```
sudo systemctl status docker --no-pager
sudo systemctl is-enabled docker
docker --version
docker compose version
cd /opt/shopsphere
pwd
cat image-tag.env
docker compose --env-file image-tag.env ps
```

Command	What it checks
systemctl status docker	Docker daemon running.
systemctl is-enabled docker	Docker configured to start at boot.
docker --version / compose version	Installed Docker and Compose versions.
pwd	Confirms you are in /opt/shopsphere.
cat image-tag.env	Shows active release tag; expected known-good tag build-24 unless you intentionally changed it.
docker compose ... ps	Shows status of five Compose services.

Expected application services: user-service, product-service, order-service, frontend and nginx. Backends should show healthy. Nginx should publish host port 80.

If services are not running, start the existing Compose stack

```
cd /opt/shopsphere
docker compose --env-file image-tag.env up -d
docker compose --env-file image-tag.env ps
```

This starts the existing stack using its current tag. Do not run pull or change image-tag.env unless you have a reason; the known-good release is build-24.

Check homepage locally on EC2

```
curl -sS -o /dev/null -w 'HTTP Status: %{http_code}\n' http://localhost/
docker compose --env-file image-tag.env logs --tail=50 nginx
```

Expected HTTP status: 200. The log command displays the last 50 Nginx log lines if the check fails.

5. Verify Node Exporter on EC2

Still inside Ubuntu EC2 SSM terminal:

```
sudo systemctl status prometheus-node-exporter --no-pager
sudo systemctl is-enabled prometheus-node-exporter
curl -fsS http://localhost:9100/metrics | head -n 10
```

Expected: Active (running), enabled, and metric text beginning with # HELP / # TYPE. A curl error 23 can appear because head closes the pipe after ten lines; the metrics endpoint may still be working.

If Node Exporter is stopped

```
sudo systemctl enable --now prometheus-node-exporter
sudo systemctl status prometheus-node-exporter --no-pager
```

This starts the installed service now and ensures it starts on future boots. Do not reinstall unless the package/service is actually missing.

6. Restart local monitoring applications on Windows

EC2 stopping does not stop the Windows applications, but closing their PowerShell windows does. Open separate PowerShell windows for Prometheus and Grafana.

Prometheus (Windows PowerShell — window 1)

```
& "C:\DevOps-Monitoring\prometheus\prometheus-3.16.0.windows-amd64\prometheus.exe"
  --config.file="C:\DevOps-Monitoring\prometheus\prometheus.yml"
  --web.listen-address="127.0.0.1:9091"
```

Starts Prometheus using the existing config. Jenkins already uses 9090, so Prometheus uses 9091. Keep this window open.

Grafana (Windows PowerShell — window 2)

```
cd C:\DevOps-Monitoring\grafana\bin
.\grafana.exe server --homepath "C:\DevOps-Monitoring\grafana"
```

Starts the extracted Grafana 13.3.1 server. Keep this window open.

Open Prometheus at http://localhost:9091 and Grafana at http://localhost:3000. Their local data/config folders remain on Windows even while EC2 is stopped.

7. Recreate the secure Node Exporter tunnel

Open a third Windows PowerShell window. Run this only after EC2 is Running and SSM is Online:

```
aws ssm start-session --target i-0af087445cc5f5878 --document-name AWS-StartPortForwardingSession
  --parameters '{"portNumber":["9100"],"localPortNumber":["9100"]}' --region ap-southeast-2
```

This forwards Windows localhost:9100 to EC2 localhost:9100 through SSM. Leave the window open. It is a live session; closing the window stops the tunnel. No public inbound rule for port 9100 is required.

Test the tunnel from another Windows PowerShell window

```
Invoke-WebRequest http://127.0.0.1:9100/metrics -UseBasicParsing | Select-Object -ExpandProperty StatusCode
(Invoke-WebRequest http://127.0.0.1:9100/metrics -UseBasicParsing).Content | Select-Object -First 10
```

Expected status code 200 and Prometheus metric text. If this fails, confirm SSM Online, Node Exporter active on EC2, and that the tunnel window still shows an active session.

8. Verify Prometheus and Grafana

Prometheus target

Open http://localhost:9091/targets. Find job ec2-node-exporter. Expected state: UP. If DOWN, check that prometheus.yml target is 127.0.0.1:9100, the tunnel is active, and Prometheus was restarted after config changes.

Prometheus query

Open http://localhost:9091/graph and query:

```
up
```

Expected value 1 for the EC2 Node Exporter target. You can also query node_memory_MemAvailable_bytes or node_filesystem_avail_bytes.

Grafana data source

Open http://localhost:3000 → Connections / Data sources → Add data source → Prometheus. Set URL to http://localhost:9091, then Save & test. Because Prometheus and Grafana both run on the same Windows laptop, localhost is correct.

If you already added this data source earlier, do not add a duplicate; use Save & test to verify it.

Dashboard

Import a Linux Node Exporter dashboard only after the target is UP and data source test succeeds. Confirm CPU, memory, disk and network panels show fresh values. Capture screenshots for the assignment.

9. Jenkins and CI/CD readiness

Jenkins runs locally on Windows and is independent of EC2. Open http://localhost:8080 and confirm the Jenkins service is running. Port 9090 was identified as occupied by Jenkins in the monitoring setup; keep Prometheus on 9091.

```
Get-Service *jenkins*
Test-NetConnection 127.0.0.1 -Port 8080
```

The first command checks the Windows Jenkins service if installed as a service. The second tests the Jenkins web port. If your Jenkins URL/port is configured differently, use the actual configured port.

Before running a pipeline, verify Docker Desktop/Linux engine is available, AWS CLI credentials/profile are valid, Jenkins credential ID aws-ecr-jenkins exists, and the EC2 instance is SSM Online. A deployment pipeline may build a new tag; the current production version remains build-24 until a successful deployment.

10. Troubleshooting

Symptom	Check / action
EC2 says Running but SSM Offline	Wait a few minutes; verify SSM Agent and instance role ShopSphere-EC2-Role with AmazonSSMManagedInstanceCore
Public IP changed	Use the new IP for browser access. SSM commands still use instance ID. Update DNS or external references if any.
Docker inactive	sudo systemctl start docker; then check docker status.
Containers stopped	cd /opt/shopsphere && docker compose --env-file image-tag.env up -d

Symptom	Check / action
A backend is unhealthy	docker compose --env-file image-tag.env logs --tail=100 user-service (replace service name as needed). Check env file a
Homepage not HTTP 200	Check Nginx and backend logs; run docker compose ps and curl localhost.
Node Exporter not running	sudo systemctl enable --now prometheus-node-exporter
Prometheus target DOWN	Check SSM port-forward window and Windows localhost:9100 metrics endpoint.
Grafana data source test fails	Confirm Prometheus UI works at localhost:9091 and data source URL is exactly http://localhost:9091.
Jenkins can't deploy	Check Jenkins credentials, IAM permissions, AWS region, SSM Online and pipeline console log.

11. Safe shutdown procedure for next time

When you finish work, stop local monitoring processes and EC2 in this order:

```
1. Save screenshots / notes.
2. Stop Grafana and Prometheus PowerShell processes with Ctrl+C.
3. Close the SSM port-forwarding session (Ctrl+C).
4. In AWS Console, select EC2 instance → Instance state → Stop instance.
   Or from Windows PowerShell:
   aws ec2 stop-instances --instance-ids i-0af087445cc5f5878 --region ap-southeast-2
   aws ec2 wait instance-stopped --instance-ids i-0af087445cc5f5878 --region ap-southeast-2
```

Do not terminate the instance. Stop is reversible; terminate is destructive. EBS storage and ECR storage may continue to incur charges while compute is stopped.

12. Quick restart checklist

Done	Action
■	Start existing EC2 instance; wait for Running and 2/2 checks.
■	Confirm SSM PingStatus Online.
■	Open SSM session.
■	Check Docker enabled/running.
■	Check image-tag.env is build-24 (or your intended release).
■	Check/start five Compose services; three backends healthy.
■	Confirm localhost HTTP 200 on EC2.
■	Confirm Node Exporter active and metrics endpoint responds.
■	Start Prometheus on Windows port 9091.
■	Start Grafana on Windows port 3000.
■	Start SSM port-forward tunnel for 9100.
■	Confirm Windows localhost:9100 responds.
■	Prometheus target ec2-node-exporter is UP.
■	Grafana Prometheus data source Save & test succeeds.
■	Check Jenkins is available before a pipeline run.

Known-good checkpoint from the last session: release build-24, five containers running, three backend services healthy, homepage HTTP 200, Node Exporter installed and active, Prometheus/Grafana installed and previously started locally. The SSM monitoring tunnel and Grafana data-source/dashboard verification were still pending.